Exact-value precision resistors are available for applications that require extremely high accuracy. Common power ratings range from 1/4 watt to 2 watts. Physical size increases with required wattage. For good stability, actual power dissipation in service should not exceed 50 per cent of the rating. Since power is dissipated in the form of heat, this is a good rule to follow, because excessive heat will decrease resistance.

Overheating may cause permanent damage to resistors. So when soldering these units in place, care must be taken to prevent these from over-heating. Excessive heating will cause discolouration of the resistor body and colour code stripes. For precision applications, a low-wattage, 1 per cent composition type made of pure carbon deposited in a spiral groove on a ceramic rod is also made.

Resistance values are generally marked on the body in English. The physical size of precision resistors may vary between manufacturers and may often be misleading as they are somewhat larger for a given rating than the common composition type. They cost several times more than ordinary-composition resistors.

As mentioned earlier, wire-wound resistors' resistance increases as these heat up. This change in resistance is quite small, but care should be taken to keep the resistor as cool as possible for resistance stability. Resistors should be mounted in a well ventilated position and should be capable of at least twice the power dissipation required. In other words, if calculations show that 5 watts will be dissipated, the resistor used should be rated at no less than 10 watts. This rule should be followed invariably, although overheating is more likely to cause permanent damage to composition resistors than wire-wound units.

PCBs and substrates. PCBs and substrates provide mechanical support for components, dissipate heat and electrically interconnect components. Above their glass transition temperature (Tg), however, organic boards have trouble performing these functions. They begin to lose mechanical strength due to resin softening and often exhibit large discontinuous changes in their out-of-plane coefficients of thermal expansion. These changes can cause de-lamination between the resin and glass fibres in the board or, more commonly, between copper traces and the resin. Furthermore, the insulation resistance of organic boards decreases significantly above Tg.

Optimisation from a heat dissipation perspective begins with the PCB. PCBs using standard FR-4 material are limited to less than 135°C temperatures, although high-temperature versions (for use up to 180°C) are available. Those manufactured using bismaleimide triazine (BT), cyanate ester (CE) or polyimide materials can be used up to 200°C or sometimes even higher temperature, with quartz–polyimide boards useful up to 260°C. Boards made with PTFE resin have Tg greater than 300°C, but are not recommended for use above 120°C due to weak adhesion of the copper layer. Use of copper improves the PCB's thermal characteristics since its thermal conductivity is more than 1000 times higher than the base FR-4 material.

Thoughtful design of the PCB along with logical placement of power-dissipating packages can result in big improvements at virtually no cost. Use of the PC board's copper to spread the heat away from the package, along with innovative use of copper mounting pads, plated through-holes and power planes can significantly reduce thermal resistance.

While ICs are getting faster and more powerful, PC boards are shrinking in size. Today's smaller PC boards (such as those found in cell phones and PDAs) and product enclosures with their higher speed demand more cooling than earlier devices. Their increased performance-to-package size ratios generate more heat, operate at higher temperatures and thus have greater thermal management requirements.

Solders. Most engineering materials are used to support mechanical loads only in applications where the use temperature in Kelvin is less than half the melting point. However, since the advent of surface mount technology, solder has been expected to provide not only electrical contact but also mechanical support at temperatures well in excess of this guideline.

In fact, at only 100°C, eutectic solder reaches a temperature over 80 per cent of its melting point and exhibits Navier–Stokes flow. Above this temperature, shear strength decreases to an unacceptable level and excessive relaxation is observed. In addition, copper-tin intermetallics can form between tin-lead solder and copper leads at elevated temperatures, which can weaken the fatigue strength of joints over time.

There are a number of solders that can be used at temperatures up to 200°C. Thus the temperature that a PWA can withstand is the lowest maximum temperature of any of the components used in the assembly of the PWA (connectors, plastic ICs, discrete components, modules, etc), the PCB and its materials, and the solder system used.

The shift to no-lead or lead-free solder as a result of the environmental and health impact of lead presents the electronics industry with reliability, manufacturability, availability and price challenges. Generally speaking, most of the proposed materials and alloys have mechanical, thermal, electrical and manufacturing properties that are inferior to lead-tin (Pb-Sn) solder and also cost more.

To date, the electronics industry has not settled on a Pb-Sn replacement. Pure tin is a serious contender to replace Pb-Sn. From a thermal viewpoint, lead-free solders require a higher reflow temperature (increasing from about 245°C for Pb-Sn to above 260°C for lead-free solder compounds) and thus greatly increase the possibility of component and PWA damage, impacting reliability.

Although the air immediately surrounding a chip will initially cool the chip's surface, that air eventually warms up and rises to the top of the chassis, where it encounters other warm air. If not ventilated, this volume of air becomes warmer and warmer, offering no avenue of escape for the heat generated by chips.

By performing thermal analy-